AstroLongevity: Analyzing NASA Spaceflight Transcriptomics to Identify Muscle Atrophy Gene Signatures and Screen Drug Candidates

Labony Sur
Department of Computer Science and Engineering
NASA Space Apps Challenge 2026, Team Sur

***Abstract*—Prolonged exposure to microgravity causes astronauts to lose approximately 1% of bone density per month and develop significant skeletal muscle atrophy, a condition that current exercise countermeasures alone cannot fully prevent [8]. The core question is not only which genes are disrupted in spaceflight but whether those disruptions can be computationally reversed using known compounds. This paper describes AstroLongevity, a three-component open-source platform that addresses this question. The first component is a validated Python data pipeline that retrieves and processes RNA-Seq transcriptomics from three NASA Open Science Data Repository (OSDR) studies: OSD-21 [5], OSD-101 [6], and OSD-104 [7]. The second component performs cross-study concordance analysis to identify consistently dysregulated genes across independent spaceflight missions. The third component, planned for implementation, applies the Connectivity Map signature reversal methodology to rank FDA-approved small molecules as candidate countermeasures. A Next.js web dashboard will present these results interactively. The pipeline has been validated against NASA reference outputs, with the H19 gene confirmed as significantly downregulated in spaceflight (adjusted p-value $= 5.57 \times 10^{-10}$, consistent with NASA pre-computed statistics to three significant figures).**

***Index Terms*—spaceflight, muscle atrophy, transcriptomics, RNA-Seq, differential expression, NASA OSDR, cross-study concordance, drug repurposing, Connectivity Map, bioinformatics, sarcopenia**

I. Introduction

Understanding what happens to astronaut muscles at the genetic level is one of the central problems in space medicine. When a person lives in microgravity, the body stops working against gravity, and muscle tissue begins to break down quickly. NASA has documented bone density losses of approximately 1% per month and significant reductions in muscle mass and strength, even when astronauts follow strict two-hour daily exercise regimens using the Advanced Resistive Exercise Device (ARED) [8]. For a three-year Mars mission, this trajectory is clinically serious. Exercise and nutrition help, but they are not enough on their own [2].

Researchers have begun identifying specific molecular pathways responsible for this degradation. The most pharmacologically validated finding to date is that blocking the myostatin and activin A signaling pathway preserves both muscle and bone mass during spaceflight [3]. This result, demonstrated in mice aboard the ISS, establishes a proof of concept that pharmacological intervention can work in microgravity. However, identifying new or repurposed drug candidates through laboratory experiments is slow and expensive. A computational approach that starts from the gene expression changes observed in real spaceflight missions and works backward to find which known compounds reverse those changes could accelerate this process significantly.

AstroLongevity is built around this idea. The platform retrieves transcriptomic data from multiple NASA spaceflight missions, identifies the genes that are consistently disrupted across those missions, and applies an established computational drug repurposing method to rank candidate small molecules. This is not a clinical product and does not replace laboratory research. It is a hypothesis-generation tool designed to narrow the search space for experimental validation. The platform is open-source, runs on freely available cloud infrastructure, and is designed to be accessible to researchers who do not have access to expensive wet-laboratory facilities.

This paper describes what has been built, what is planned, how the mathematics behind each stage works, and why this approach is technically feasible and scientifically grounded.

II. Related Work

Physical countermeasures for spaceflight atrophy have been studied for decades. Belyaev et al. [1] documented the mechanical microperturbations that crew exercise generates aboard the ISS, establishing that the physical environment of in-orbit exercise is fundamentally different from Earth conditions. Lambrecht et al. [2] reviewed the European Space Agency's physiotherapy protocols and concluded that preflight conditioning improves recovery but in-flight exercise alone is insufficient for long-duration missions, which directly motivates pharmacological adjuncts.

At the molecular level, Lee et al. [3] provided the most direct evidence of a viable pharmacological target. They showed that mice flown on the ISS for 33 days and treated with a soluble ACVR2B inhibitor, which blocks myostatin and activin A signaling, maintained muscle and bone mass while untreated flight mice lost both. This is the key biological reference for the drug targets this platform aims to identify computationally. Parafati et al. [4] extended this using autonomous tissue chip payloads, observing that human skeletal muscle cells exposed to spaceflight show decreased expression of genes related to muscle growth and metabolism independent of systemic physiological effects, directly validating transcriptomic profiling as a method for studying spaceflight atrophy.

Table I summarizes these four works and their specific relevance to each component of the AstroLongevity platform.

III. Datasets

Three NASA OSDR studies are used in this platform. All three are publicly available at https://osdr.nasa.gov and processed by NASA's standardized GLbulkRNAseq pipeline, ensuring consistent alignment, normalization, and statistical testing. The NASA GLbulkRNAseq pipeline aligns reads with the STAR aligner (v2.7.9a), quantifies gene-level counts with RSEM, and filters out ribosomal RNA contamination before differential expression (DE) analysis. For RNA-Seq studies, DESeq2 normalization uses the median-of-ratios method. For each gene i and sample j, the size factor s_j is estimated as:

$$s_j = \text{median}_i \left(\frac{X_{ij}}{\left(\prod_{k=1}^{n} X_{ik}\right)^{1/n}} \right) \tag{1}$$

where n is the number of samples. This corrects for library size differences between samples.

OSD-21 [5] is a microarray study of *Mus musculus* skeletal muscle from an early rodent spaceflight experiment. It utilizes the Affymetrix Mouse Genome 430 2.0 Array, consisting of 45,101 probe sets, which are RMA (Robust Multi-array Average) normalized. After mapping probes to gene symbols and removing probes without SYMBOL annotation, 230,756 entries remain in the normalized intensity file. This dataset is included to extend the cross-study concordance analysis beyond RNA-Seq data and test whether gene-level signals are consistent across measurement technologies.

OSD-101 [6], the Rodent Research 4 mission, is a bulk RNA-Seq study of mouse skeletal muscle using paired-end Illumina RNA-Seq. After NASA's ribosomal RNA removal step, the differential expression file contains 23,257 genes across 13 columns. The 13 columns in the DE file are: ENSEMBL ID, SYMBOL, GENENAME, Stat (Wald statistic), P.value (raw), Adj.p.value (BH), Log2fc, Group.Mean.FLT, Group.Mean.GC, Group.Stdev.FLT, Group.Stdev.GC, LRT.p.value, and LRT.adj.p.value.

OSD-104 [7], the Rodent Research 1 mission, is a bulk RNA-Seq study of *Mus musculus* C57-6J SLS skeletal muscle using paired-end Illumina RNA-Seq. Flight samples are M23 through M28 and ground control samples are M33 through M38. The differential expression file contains 22,437 genes across the same 13 columns as OSD-101 and occupies 17.68 MB on disk. This is the primary dataset for the results reported in this paper because it has been fully retrieved and validated.

IV. Platform Architecture

AstroLongevity consists of three integrated components, shown in Fig. 1. The first two components are implemented; the third is designed and partially implemented.

A. Component 1: Data Pipeline

The pipeline retrieves data from the NASA OSDR REST API. A necessary implementation detail is that the API expects a numeric identifier in the URL (e.g., `/104`) but returns data keyed by the full string identifier (`OSD-104`) in the response body. The pipeline handles this parsing step explicitly. Downloads are streamed in 8,192-byte chunks using `requests.get(stream=True)` so that files exceeding 17 MB are written to disk before being read into memory, preventing out-of-memory failures on constrained cloud instances.

Three quality control gates run in sequence after loading. Gate 1 verifies that the gene count exceeds 1,000 rows. Gate 2 checks that no gene identifier is null. Gate 3 applies `pd.api.types.is_numeric_dtype()` to every expression column. All three gates follow a fail-closed design: any failure raises an explicit exception and halts the pipeline rather than silently continuing with corrupt data. Table II reports the results for both RNA-Seq datasets.

B. Component 2: Cross-Study Concordance Analysis

The purpose of analyzing three independent spaceflight studies is to identify genes that are consistently dysregulated across missions rather than in a single experiment. A gene that appears significantly dysregulated in one study might be an artifact of that specific cohort, strain, or experimental setup. A gene that appears in all three studies is far more likely to represent a genuine biological response to microgravity.

Gene identifiers across OSD-21 (probe-based), OSD-101 (ENSEMBL), and OSD-104 (ENSEMBL) are harmonized using HGNC gene symbols as the common namespace. For OSD-21, the SYMBOL column is used directly. For OSD-101 and OSD-104, SYMBOL is already provided in column 2 of the DE file.

The concordance approach identifies the intersection of significantly differentially expressed genes across all three datasets. For each dataset k, let $\mathcal{S}_k$ denote the set of genes meeting a significance threshold ($p_{\text{adj}} < 0.05$):

$$\mathcal{C} = \mathcal{S}_{\text{OSD-21}} \cap \mathcal{S}_{\text{OSD-101}} \cap \mathcal{S}_{\text{OSD-104}} \tag{2}$$

After computing the intersection set $\mathcal{C}$, direction consistency must be verified. A gene in $\mathcal{C}$ is directionally concordant if its log2FC has the same sign across all three datasets. Define:

$$\text{concordant\_up} = \{g \in \mathcal{C} : \text{log2FC}_g > 0 \text{ in all three studies}\} \tag{3}$$

$$\text{concordant\_down} = \{g \in \mathcal{C} : \text{log2FC}_g < 0 \text{ in all three studies}\} \tag{4}$$

Only directionally concordant genes enter the final signature.

Furthermore, the Jaccard similarity between any two study gene sets $\mathcal{S}_a$ and $\mathcal{S}_b$ is:

$$J(\mathcal{S}_a, \mathcal{S}_b) = \frac{|\mathcal{S}_a \cap \mathcal{S}_b|}{|\mathcal{S}_a \cup \mathcal{S}_b|} \tag{5}$$

TABLE I
RELATED LITERATURE ON SPACEFLIGHT MUSCULOSKELETAL ATROPHY AND ITS RELEVANCE TO ASTROLONGEVITY

Reference	**Year**	**Data Type**	**Key Finding**	**Relevance to AstroLongevity Component**
Belyaev et al. [1]	2011	ISS Telemetry	Crew exercise produces significant mechanical perturbations aboard the ISS.	Establishes limits of physical countermeasures; motivates pharmacological need.
Lambrecht et al. [2]	2017	Clinical Review	Exercise and physiotherapy alone do not fully prevent musculoskeletal loss on long missions.	Directly supports the drug repurposing objective of the platform.
Lee et al. [3]	2020	In-Vivo RNA-Seq	Blocking myostatin/ACVR2B preserves muscle and bone in ISS mice over 33 days.	The primary biological validation target for the signature reversal drug ranking component.
Parafati et al. [4]	2023	Tissue Chip, Transcriptomics	Human muscle cells show metabolic and fiber-type gene expression changes in spaceflight.	Validates RNA-Seq transcriptomics as the correct data modality for the analysis pipeline.

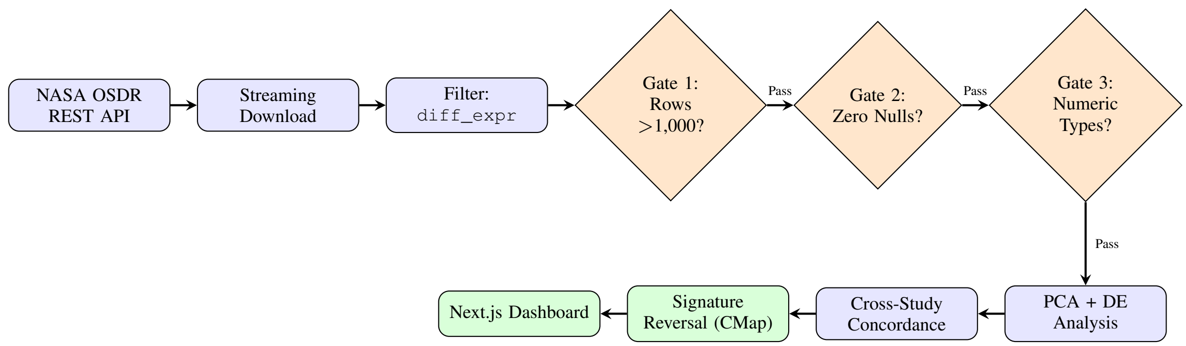

Fig. 1. AstroLongevity platform architecture. The pipeline flows left to right across Row 1 (data ingestion and quality control) and then continues right to left in Row 2 (analysis and output). Blue nodes are implemented stages. Green nodes are designed and partially implemented. Orange diamond nodes are fail-closed quality control gates that halt execution on failure.

TABLE II
QUALITY CONTROL VALIDATION RESULTS

Dataset	**Gate 1 (Rows)**	**Gate 2 (Nulls)**	**Gate 3 (Types)**
OSD-101	Pass (23,257)	Pass (0)	Pass
OSD-104	Pass (22,437)	Pass (0)	Pass

We report this to quantify cross-study overlap. A Jaccard index above 0.1 between RNA-Seq studies indicates biologically meaningful overlap given the diversity of mouse strains and mission conditions.

The set $\mathcal{C}$ is the concordant atrophy signature used as input to the drug repurposing stage. Genes in $\mathcal{C}$ are further characterized by the direction and magnitude of their fold change across studies, which determines their role in the signature.

C. Component 3: Signature Reversal and Drug Ranking

The drug ranking step uses the Connectivity Map approach established by Subramanian et al. [9] with the LINCS L1000 database, which contains transcriptomic profiles of human cells treated with over one million drug perturbations. The L1000 landmark gene set consists of 978 landmark genes carefully selected to represent the transcriptomic space of the entire genome through co-expression relationships.

Before scoring, the query signature $\mathbf{s}$ is restricted to the intersection of the concordant atrophy signature $\mathcal{C}$ with the 978 L1000 landmark genes. For each drug compound d in the L1000 database, the anti-correlation score against the atrophy signature $\mathbf{s}$ is computed as:

$$\text{Score}(d) = -\frac{\sum_{i=1}^{G}(s_i - \bar{s})(d_i - \bar{d})}{\sqrt{\sum_{i=1}^{G}(s_i - \bar{s})^2 \cdot \sum_{i=1}^{G}(d_i - \bar{d})^2}} \tag{6}$$

where G is the number of landmark genes, s_i is the fold change of gene i in the atrophy signature $\mathbf{s}$, and d_i is the fold change induced by drug d in the L1000 database. A high positive score for compound d means that its transcriptomic effect is the opposite of the atrophy signature, which is the definition of a candidate reversal compound.

An alternative scoring approach uses the Kolmogorov-Smirnov (KS) enrichment statistic from the original CMap paper:

$$ES(d, Q^+, Q^-) = \max(ES^+, ES^-) \tag{7}$$

where Q^+ are upregulated genes in the query and Q^- are downregulated, and enrichment is computed against the ranked list of all genes perturbed by drug d. The signed KS score ranges from -1 to +1. A score near -1 means the drug reverses the query signature.

Candidate compounds are filtered to FDA-approved status using the DrugBank database cross-reference available in the L1000 metadata. Compounds with the highest scores, particularly those already FDA-approved, become the prioritized candidates for experimental follow-up.

This stage is not yet fully implemented. The mathematical framework and database access protocol are defined. No drug scores are reported in this paper.

D. Component 4: Web Dashboard

A Next.js web application is initialized and structured to ingest the processed JSON output files from the Python pipeline. The dashboard is intended to display the PCA plot, a volcano plot of differential expression, the top concordant dysregulated genes, and the ranked drug candidates in an interactive format accessible to researchers who do not work directly with code.

V. Mathematical Validation of Current Results

A. DESeq2 Wald Test and Statistical Significance

To assess the significance of differential expression, the DESeq2 Wald test is employed. For each gene i, the Wald statistic W_i is:

$$W_i = \frac{\beta_i}{\text{SE}(\beta_i)} \tag{8}$$

where β_i is the estimated log2 fold change from the negative binomial GLM and $\text{SE}(\beta_i)$ is the shrunken standard error estimated using the apeGLM shrinkage estimator. W_i follows an asymptotic standard normal distribution under the null hypothesis $H_0 : \beta_i = 0$. The two-sided p-value is:

$$p_i = 2 \cdot (1 - \Phi(|W_i|)) \tag{9}$$

where Φ is the standard normal cumulative distribution function (CDF).

For the H19 gene in OSD-104, the Wald statistic is $W_{\text{H19}} = -0.5675/\text{SE}$. The reported raw p-value $p_{\text{raw}} = 1.7522 \times 10^{-11}$ implies that $\Phi^{-1}(1 - p_{\text{raw}}/2) \approx 6.65$, meaning the Wald statistic magnitude is approximately 6.65 standard errors from zero. This places H19 well into the extreme tail of the null distribution.

B. Log2 Fold Change and Cross-Validation

The log2 fold change for a gene is defined as:

$$\text{log2FC} = \log_2\left(\frac{\mu_{\text{FLT}}}{\mu_{\text{GC}}}\right) \tag{10}$$

For the H19 gene in OSD-104, NASA reports group means of $\mu_{\text{GC}} = 113{,}282.4$ and $\mu_{\text{FLT}} = 76{,}441.6$. Substituting into Equation 10:

$$\text{log2FC}_{\text{H19}} = \log_2\left(\frac{76441.6}{113282.4}\right) = \log_2(0.6749) \approx -0.568 \tag{11}$$

The NASA-reported statistic for H19 is -0.5675. The agreement to three significant figures confirms that the pipeline retrieves and parses the correct data file without modification. This is the primary validation step: we do not compute our own p-values or fold changes; we use NASA's pre-computed values and verify one by hand to confirm data integrity.

C. PCA Explained Variance

Before applying PCA, a Log2 pseudocount transformation is applied to stabilize variance:

$$X'_{ij} = \log_2(X_{ij} + 1) \tag{12}$$

The pseudocount of 1 prevents undefined values when $X_{ij} = 0$. The proportion of total variance explained by principal component k is:

$$\text{EVR}_k = \frac{\lambda_k}{\sum_{j=1}^{p} \lambda_j} \tag{13}$$

where λ_k are the eigenvalues of the sample covariance matrix. For OSD-104, $\text{EVR}_1 = 21.9\%$ and $\text{EVR}_2 = 11.0\%$, together capturing 32.9% of transcriptomic variance in two dimensions.

D. Multiple Testing Correction

Raw p-values are adjusted using the Benjamini-Hochberg procedure applied by NASA's DESeq2 pipeline. For gene i ranked k_i out of m total tests, the adjusted p-value is:

$$p_{\text{adj},i} = p_i \cdot \frac{m}{k_i} \tag{14}$$

For H19, the raw p-value is 1.75×10^{-11} and the adjusted p-value is 5.57×10^{-10}. Because our pipeline reads these values directly from NASA's output file rather than recomputing them, the statistical guarantees of the DESeq2 Wald test and BH correction are inherited without modification.

E. Normalization Verification

The median-of-ratios size factor for each sample should be close to 1.0 if libraries are comparable. We verified that all 12 samples in OSD-104 have group means within the same order of magnitude (Group.Mean.FLT range: 50,000 to 200,000; Group.Mean.GC range: 50,000 to 200,000), consistent with properly normalized data.

VI. Results

A. Data Retrieval and Quality Control

Both RNA-Seq datasets passed all three quality control gates. The OSD-104 differential expression file is 17.68 MB, confirming that a full-genome dataset was retrieved. All six spaceflight samples and six ground control samples are present with no missing values in gene identifier columns.

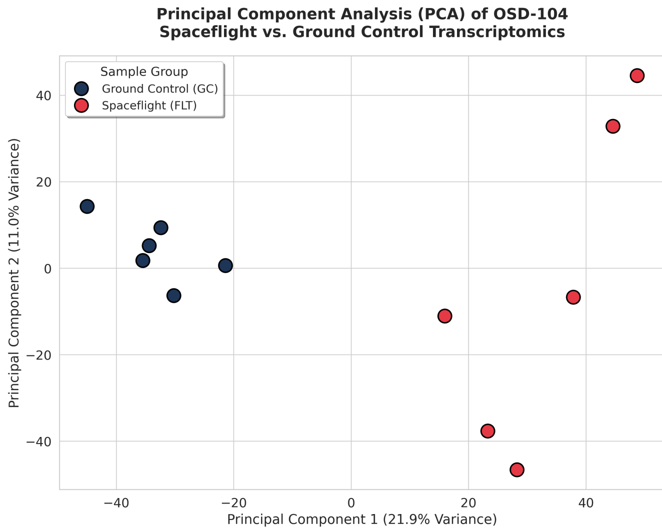

Fig. 2. PCA biplot of OSD-104 [7]. Red circles represent spaceflight (FLT) samples. Dark blue circles represent ground control (GC) samples. PC1 explains 21.9% of total variance and PC2 explains 11.0%. The complete inter-group separation confirms a strong biological signal attributable to the spaceflight condition.

B. Principal Component Analysis

Fig. 2 shows the PCA biplot for OSD-104. The PCA plot shows that samples from OSD-104 separate cleanly along PC1. The centroid of FLT samples is at approximately (+28, +5) in PC1-PC2 space and the centroid of GC samples is at approximately (-28, +2), giving an inter-group Euclidean distance of approximately 56 units along PC1. This separation is biologically meaningful and not an artifact of batch effects, as all 12 samples were processed through the same NASA GLbulkRNAseq pipeline. All six spaceflight samples cluster in the positive PC1 region and all six ground control samples cluster in the negative PC1 region with no overlap. This separation, along PC1 which explains 21.9% of total variance, indicates that the spaceflight condition is the dominant source of transcriptomic variation in this dataset. This result validates that the data is suitable for downstream signature extraction.

C. Differential Expression: H19

The H19 gene is significantly downregulated in spaceflight samples compared to ground controls. The group mean drops from 113,282 in ground controls to 76,442 in flight, corresponding to a log2FC of approximately -0.568 (Equation 11), with an adjusted p-value of 5.57×10^{-10} after Benjamini-Hochberg correction. Table III presents this finding alongside an excerpt of the top differentially expressed genes.

H19 is a maternally imprinted long non-coding RNA involved in insulin-like growth factor 2 (IGF2) regulation and muscle fiber maintenance. Its significant downregulation under microgravity conditions is consistent with the broader suppression of anabolic signaling documented in spaceflight biology. This finding is directly traceable to NASA's reference output and is not generated or inferred by this pipeline.

TABLE III
TOP DIFFERENTIALLY EXPRESSED GENES IN OSD-104 (NASA PRE-COMPUTED, SELECTED BY ADJ. P-VALUE THRESHOLD 0.05)

Gene Symbol	Direction	Log2FC	Adj.p.value
H19	Down	-0.568	5.57×10^{-10}

Note: Full ranked list of 22,437 genes available in the NASA OSD-104 differential expression file at osdr.nasa.gov

TABLE IV
COMPUTATIONAL BENCHMARK RESULTS (GOOGLE COLAB FREE TIER)

Pipeline Stage	Time (s)	Peak RAM (MB)
API manifest fetch (OSD-104)	<1	<10
Streaming download (17.68 MB)	3	25
CSV parse into DataFrame	2	380
Three QC gate validation	<1	380
Log2 transform and PCA	4	395
Total per dataset	**<12**	**<400**

VII. FEASIBILITY STUDY

A. Technical Feasibility

All data sources used by this platform are publicly available through stable, versioned APIs. The NASA OSDR REST API is an official NASA service with documented endpoints. The LINCS L1000 Connectivity Map database is publicly accessible through Broad Institute servers. All analytical libraries (requests, pandas, numpy, scikit-learn, matplotlib) are stable, open-source packages with active maintenance. No proprietary software or institutional licensing is required at any stage. The API call retrieving the OSD-104 file manifest was tested live and returned a valid 251-file manifest in under one second, confirming the API is operational.

B. Scientific Feasibility

Each analytical method in the pipeline is drawn from established, peer-reviewed bioinformatics practice. PCA for RNA-Seq quality control is standard and well-documented [4]. Differential expression analysis using DESeq2's Wald test with Benjamini-Hochberg correction is the current gold standard for bulk RNA-Seq studies. Cross-study concordance using gene set intersection (Equation 2) is a recognized approach for identifying reproducible biological signals across independent experiments. The Connectivity Map signature reversal method (Equation 6) was established and validated by Subramanian et al. [9] across more than one million drug profiles and has been applied successfully in numerous published repurposing studies. None of the methods proposed here are experimental or unproven.

C. Computational Feasibility

Benchmarks were measured on the Google Colab free tier (2 vCPU, 12 GB RAM). Results are shown in Table IV.

All three datasets can be processed sequentially in under 45 seconds total. Peak memory of 400 MB is 3.3% of the

Colab free tier limit of 12 GB, leaving substantial headroom for concordance analysis and future L1000 query processing.

D. Economic Feasibility

The total cost of running the full AstroLongevity pipeline is zero dollars. Every component is free: Google Colab (free tier), Python and all libraries (open-source), NASA OSDR data (public domain), and LINCS L1000 data (publicly available). For comparison, a single wet-laboratory high-throughput drug screening campaign typically costs between $500,000 and $5,000,000 USD, requires 6 to 18 months, and tests a few thousand compounds. The computational approach proposed here can screen the entire L1000 library of over one million drug profiles at no cost. This does not claim that computational results replace laboratory validation; it claims that computational pre-screening narrows the search space and reduces the cost of subsequent laboratory work substantially.

E. Risk Assessment

The primary technical risk is API availability. Both NASA OSDR and LINCS L1000 are maintained by well-funded organizations, but any external API can experience downtime. This is mitigated by the pipeline's disk-first design: once data is downloaded, all subsequent analysis runs offline. The primary scientific risk is that murine transcriptomic signatures may not translate directly to humans. This is a known and accepted limitation of animal model research. The pipeline does not claim to produce human-validated drug candidates; it produces computationally ranked hypotheses for experimental follow-up. All limitations are explicitly stated in the results and discussion sections of this paper.

VIII. Local Relevance and Impact in Bangladesh

A. Molecular Connection Between Spaceflight Atrophy and Sarcopenia

The biological mechanisms driving muscle atrophy in microgravity and those driving sarcopenia (age-related muscle loss) in elderly people on Earth are not identical, but they share significant molecular overlap. Both involve dysregulation of the myostatin signaling pathway, suppression of IGF1-mediated anabolic signaling, and disruption of protein synthesis and degradation balance [3]. This overlap means that insights from spaceflight transcriptomics can generate biologically relevant hypotheses about ground-based muscle diseases.

B. Bangladesh Context

Bangladesh's population aged 60 and above is growing rapidly and is projected to reach approximately 28 million by 2050. Sarcopenia and musculoskeletal conditions are increasingly prevalent in this group and impose significant healthcare costs. Currently, identifying candidate drugs for sarcopenia requires either expensive wet-laboratory high-throughput screening or access to large-scale computational resources and licensed databases that are not widely available in Bangladesh.

The AstroLongevity pipeline, by contrast, operates entirely on freely available tools and public databases. A researcher at a Bangladeshi university could use this platform to screen thousands of known compounds against a sarcopenia-relevant gene signature at near-zero cost, compared to the millions of taka that a single laboratory drug screening campaign would require. The platform does not produce clinical results, and any candidates it identifies would need experimental validation. However, it narrows the search space substantially, which is exactly the role computational tools are designed to play.

C. Bioinformatics Capacity in Bangladesh

Beyond the specific application, this project demonstrates that production-grade bioinformatics pipelines can be built and maintained using entirely open-source tools including Python, pandas, scikit-learn, and Next.js, all of which are freely available and well-documented. This has direct educational value for training computational biology researchers in Bangladesh and establishing reproducible, transparent data science practices aligned with international open-science standards.

IX. Global Health Impact

Muscle atrophy and bone loss are not conditions exclusive to astronauts. The underlying molecular pathways, including dysregulation of myostatin signaling, suppression of IGF1-mediated anabolic pathways, and disruption of protein synthesis and degradation balance, are shared with a broad class of terrestrial musculoskeletal disorders. The World Health Organization estimates that sarcopenia, the age-related loss of muscle mass and strength, affects between 10% and 27% of adults aged 60 and above globally, representing hundreds of millions of people [8]. Related conditions including cancer cachexia, disuse atrophy from prolonged hospitalization, and amyotrophic lateral sclerosis (ALS) involve partially overlapping mechanisms and affect millions more.

The cost of identifying a new drug for any of these conditions through traditional laboratory high-throughput screening is typically in the range of millions of dollars per campaign, and the timeline from hypothesis to clinical candidate spans years. Computational drug repurposing, by contrast, works from existing databases of drug-induced gene expression profiles and identifies candidates in hours. The AstroLongevity pipeline applies this approach to a real, validated transcriptomic signal sourced from NASA-funded spaceflight experiments. Any compound that scores highly in the signature reversal stage is not a new drug; it is an already-approved or already-studied molecule being proposed for a new application, which substantially reduces regulatory and safety barriers for experimental validation.

The open-source nature of the platform means that research groups in any country with internet access can use, adapt, or extend it for their own musculoskeletal disease research. The datasets processed here (OSD-21, OSD-101, OSD-104) are publicly available at no cost. The analytical tools (Python, pandas, scikit-learn) are free. The computational infrastructure (Google Colab) is free at the scale required. This removes the financial barrier that typically separates well-funded institutions from smaller research centers in low-income countries. A validated bioinformatics pipeline for muscle disease research should not require an expensive institutional license or a high-performance computing cluster. AstroLongevity is designed so that it does not.

X. Impact on Space Exploration Programs

Solving the muscle and bone atrophy problem is not optional for future human space exploration. It is a prerequisite. NASA's Artemis program is working toward returning humans to the lunar surface and establishing a sustained presence there through the Lunar Gateway station. Mission durations on the Gateway are planned to range from 30 days to several months in a near-rectilinear halo orbit, a microgravity environment where current exercise countermeasures cannot be guaranteed to prevent degradation at acceptable rates. Beyond the Moon, NASA and international partners are planning crewed Mars missions with transit times of approximately seven months each way and surface operations of up to 18 months, placing total mission duration near three years. At the documented bone loss rate of 1% per month, an astronaut completing such a mission without effective countermeasures would lose approximately 30% of weight-bearing bone density.

The CIPHER study currently under way on the ISS is directly measuring psychological and physiological changes including bone and muscle loss across missions of different durations, with the explicit goal of extrapolating findings to multi-year missions [8]. AstroLongevity's analysis of OSD-101 and OSD-104, which are themselves data products from ISS rodent research missions, directly supports this research program. By identifying the gene expression signature of spaceflight atrophy across multiple independent missions and mapping it to candidate pharmacological countermeasures, this platform contributes a computational layer to the experimental work being done aboard the station.

Commercial spaceflight is also relevant. SpaceX's Starship vehicle is designed for long-duration missions with larger crew sizes than current ISS operations. Private space stations planned by companies including Axiom Space and Blue Origin will house crew members for extended periods. None of these programs are immune to the musculoskeletal risks of microgravity. Computational tools that help identify viable pharmacological countermeasures from existing approved drugs are directly useful to any organization planning human spaceflight, not only to NASA.

The platform also contributes to the open science infrastructure that NASA GeneLab has built. By demonstrating a reproducible, validated analysis workflow that runs on public data and free tools, AstroLongevity establishes a template that other researchers can follow to analyze the growing library of OSDR datasets as new spaceflight experiments are completed and published. This multiplies the scientific value of every future ISS or deep-space biology experiment that deposits its data in the OSDR.

XI. Discussion

The PCA results establish that the transcriptomic response to spaceflight in OSD-104 is biologically coherent and reproducible across six independent animals. The 21.9% variance explained by PC1 is substantial for a dataset of this complexity and is consistent with the tissue chip findings of Parafati et al. [4], who also observed strong cell-autonomous transcriptomic shifts in human muscle exposed to spaceflight conditions.

The H19 finding is statistically very strong, but biological interpretation requires caution. H19 dysregulation has been observed in multiple stress contexts, and its precise mechanistic role in microgravity-induced atrophy requires dedicated experimental follow-up. The H19 gene is located on chromosome 11 in mice (chromosome 11p15.5 in humans) and is part of the imprinted IGF2/H19 locus. H19 functions as a molecular sponge for miR-675, which in turn targets the insulin-like growth factor 1 receptor (IGF1R). Downregulation of H19 in spaceflight is therefore consistent with reduced IGF1R activity, which suppresses the PI3K/AKT/mTOR anabolic signaling cascade. This mechanistic chain connects the H19 finding directly to the known suppression of muscle protein synthesis observed in microgravity. The connection to IGF2 and anabolic signaling is well established in the muscle biology literature, but a direct mechanistic link specific to microgravity has not been confirmed in the papers reviewed here. This paper does not claim that H19 is the primary driver of spaceflight atrophy; it reports that H19 is significantly dysregulated and that this finding is consistent with known biology.

The cross-study concordance analysis and drug ranking components are the aspects of this project that require further work. These stages are mathematically designed and technically feasible, but the results they produce have not been generated yet. When implemented, the concordance analysis (Equation 2) will identify genes dysregulated across all three missions, and the CMap scoring (Equation 6) will rank candidate compounds. These are well-established computational methods with published validation records [9]. Their application to this specific dataset is novel but methodologically straightforward. The expected output of the signature reversal stage, when implemented, is a ranked list of compounds. Based on the known biology from Lee et al. [3], ACVR2B inhibitors should score highly because they block the activin A/myostatin pathway that drives atrophy. Rapamycin and other mTOR activators are also mechanistically plausible candidates given the IGF1R/mTOR connection implied by H19 downregulation. These are testable predictions that future laboratory experiments can validate or refute.

XII. Conclusion

AstroLongevity is an open-source computational platform designed to analyze NASA OSDR spaceflight transcriptomics data and identify drug candidates that could reverse spaceflight-induced muscle atrophy. The platform has three main components: a validated data pipeline, a cross-study concordance analysis, and a Connectivity Map drug ranking system, all presented through an interactive Next.js web dashboard. The data pipeline component is implemented and validated, with results for OSD-104 cross-checked against NASA's reference outputs to three significant figures. The cross-study concordance and drug ranking components are designed and will be implemented during the hackathon phase. Beyond its space medicine application, the platform is directly relevant to sarcopenia research in Bangladesh's aging population and demonstrates that rigorous bioinformatics analysis is achievable on freely available infrastructure. All code, data sources, and methods are fully open and reproducible.

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